Importance of Memory for an IoT Device

There are three parallel verticals which IoT works on:

Embedded design. Most IoT devices are handheld and low-powered where data speed impacts both the hardware and the information going to the cloud. So effective data speed and even device power is required.

Mobile apps. These incorporate edge computing for storage of less complex data in the phone itself.

Cloud competencies. On the cloud side, definitely a lot of security is required. For data centres, security should impact data storage within memory in a big way. Solutions like cryptography can work for both embedded as well as for the cloud using memory pipelining.

In all these scenarios, memory is highly required.

The future is about AI happening completely on the edge where both the processor and the memory need to be efficient. "They are like husband and wife. One cannot go too fast, otherwise the marriage will break. Everything has to happen in tandem," explains Nikhil.

Convolutional neural networks (CNNs) do a lot of processing via multiply-accumulate (MAC) operations (which is to multiply two values and store the result in the accumulator). Here the value has to be held for a certain time while the next operation is being performed. But if the time taken to deliver and receive that value is more than its processing speed, then there is a problem. One solution is to bring memory closer to a system, so that with the interface pipelines the bandwidth speed can be increased. This is highly required as we increasingly move towards edge computing.

Memory from the IoT perspective

IoT generally involves communicating, processing, and storing data. And although microcontrollers can effectively handle embedded systems, they cannot handle high-speed data control, transfer, and storage required for communication with the cloud. Therefore, memory architecture needs to be strengthened for IoT. "An embedded system is required that has low limitations with respect to size, data as well as power," says Dharmendra Kumar, IoT & Digital Business Head at Arihant.

He strongly recommends design engineers to go for such memories as EMC and universal flash storage (UFS), which have both microcontroller and intelligence inside them. "Memory from an IoT point of view requires the presence of intelligence not just for the software but also for the hardware," he emphasises.

In recent years, mobile phones have emerged as powerful tools for enabling seamless functioning of IoT devices. So, if a simple operation such as switching on/off an IoT device is required, the related data can then be stored on the device itself without memory. But if a complex process such as data intelligence and data acquisition from the sensor needs to be carried out, then memory is required.

At the same time, some cases need network considerations as well. Take for example narrowband IoT (NB-IoT). It is a part of the GSM technology in which data is in a constant state of motion. So why not eliminate the need for external memories and simply place the memory within the NB-IoT module? For that to happen, the GSM platform as well as the embedded system can be interfaced via the concept of host microcontroller. How much data is suitable for traveling from hardware to software in an embedded system also requires careful consideration.

Memory for mobile robotics

The efficiency of a robot determines the quality of the work output, for which there should be intelligent processing and controlling. To achieve this, memory management and organisation is important. A communication interface between the operating system and the embedded software is needed for effective synchronisation between two or more different platforms.

In an industry, typically there are multiple workstations, each having different needs. For example, one workstation calls for materials from stores, another for transportation of finished goods to a warehouse, and so on. Here, the industrial mobile robot employed to carry out such tasks should be able to do so effectively from one workstation to another.

Since it is difficult to predict the nature and application of the calls in such a scenario, these should be prioritised based on risk and urgency, which requires intelligence in the robot. If the task is of high risk or high urgency, then it should be given the first priority.

This assigning of priorities is similar to weighing of the neural network systems. Weights are assigned here in the form of call requests. Other than that, single-board computers can be employed for queue mechanism for which a stack can be present to perform the memory organisation and call request management. There should also be a mobile robot localisation part to implement locomotion for the robot movement.

"Localisation means that there should be an obstructionless path for the robot to move. Robot condition is also important to keep in check the

Power versus Data Tradeoff?

Earlier, while implementing AI, not much thought was given regarding power consumption, which has created challenges today with respect to efficient power handling. To overcome this, a complete system realignment needs to be done so that there is no tradeoff any more between power and data.

Regarding system realignment, there is a need for AI to be low powered even when handling a huge amount of data. This can be achieved by realigning our operations. May be there will be a RAM which. while storing data, is intelligent enough to handle information when requested.

Concerning power and data, one thing is for sure: We cannot continue to consume power the way we are consuming. Power is critically important going forward.